

Risk-Sensitive Stochastic Model Predictive Control (SMPC) for Autonomous Rover Navigation in High-Variance Terrain

Team TerraRisk

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Abstract

Autonomous rovers operating on unstructured, deformable terrain, planetary regolith, sand, or mud, must plan trajectories under substantial and often heavy-tailed uncertainty in soil mechanics. Standard deterministic Model Predictive Control (MPC) and Gaussian Stochastic MPC (SMPC) either react too conservatively to unknown terrain or under-weight rare but catastrophic entrapment events caused by hidden pockets of highly compressible soil. Existing risk-aware navigation methods largely infer this risk from exteroceptive, geometry-based sensing (LiDAR and camera imagery of elevation, roughness, or slope), which cannot distinguish visually similar terrain patches that differ sharply in underlying soil strength. This project formulates a Conditional Value-at-Risk (CVaR)-based SMPC controller that supplements standard LiDAR/camera perception with direct physical measurements of the rover's own interaction with the ground, wheel slip, sinkage depth, drive-motor torque, and body attitude, feeding a Bekker-Wong terramechanics model so that risk is evaluated from the soil's actual mechanical behaviour rather than its appearance alone. The controller is benchmarked in simulation against deterministic NMPC and Gaussian SMPC baselines on entrapment rate and path efficiency.

1. Engineering System

The engineering system under study is a class of Off-Road Unmanned Ground Vehicles (UGVs), with particular emphasis on planetary exploration rovers. The scope is narrowed to the independent wheel-drive and steering control architecture of a skid-steer or Ackermann-type rover operating on unstructured, deformable ground where the vehicle-terrain interaction model, captured through classical Bekker-Wong terramechanics, is only known probabilistically. The controller sits at the trajectory-planning and low-level actuation interface: it consumes probabilistic terrain estimates and produces steering-angle and wheel-torque/velocity commands over a finite receding horizon. Unlike prior systems that derive these terrain estimates purely from exteroceptive sensing (LiDAR/camera), the rover here is additionally instrumented to sense its own physical interaction with the ground, as detailed in Section 3.

2. Problem Statement

In off-road and extraterrestrial navigation, terrain information is typically available only as probabilistic estimates of soil cohesion and internal friction angle, obtained from orbital imagery or forward-facing sensors. Classical MPC and even standard Gaussian SMPC formulations assume light-tailed (typically Gaussian) uncertainty and enforce simple chance constraints on state variables. Real terrain hazards, however, such as isolated pockets of

highly compressible soil hidden beneath a visually uniform surface, exhibit heavy-tailed statistics: rare but severe events dominate risk.

A further, related limitation runs through most existing risk-aware off-road planners, including the closest prior work identified in Section 4.3: their risk estimate is built almost entirely from LiDAR and camera data, that is, from the geometry and appearance of the terrain (elevation, roughness, slope). These sensors answer “what does the ground look like,” not “how will the ground actually behave under my wheel.” Two patches of terrain can be geometrically and visually near-identical, flat, uniform sand, while one is firmly packed and fully traversable and the other is loosely packed and capable of causing entrapment. A purely LiDAR/camera-based risk map cannot tell these apart, which forces a choice between two failure modes: treat all visually-ambiguous terrain as safe (under-reacts to real tail risk) or treat all of it as hazardous (over-reacts, discards otherwise-traversable routes, and needlessly restricts the paths the rover is willing to explore).

When a rover enters an undetected hazard zone, excessive wheel slip drives sinkage past a recoverable threshold, resulting in irreversible entrapment and mission loss. The central control-theoretic problem addressed by this project is therefore twofold: (i) to ground terrain-risk estimation in the physical soil-mechanical quantities that actually govern sinkage, rather than in geometric/visual proxies alone, and (ii) to design a trajectory-planning and control law that explicitly bounds the worst-case (tail) risk of catastrophic sinkage arising from that estimate, rather than only its expected value, while remaining computationally tractable within a receding-horizon framework and respecting hard actuator limits such as motor torque bounds.

3. Objectives

- Develop a stochastic vehicle-terrain model: formulate a simplified rover dynamic model with probabilistic wheel-slip and sinkage states, parameterised by uncertain soil cohesion and internal friction angle following Bekker-Wong terramechanics.
- Extend terrain sensing beyond LiDAR/camera: in addition to standard exteroceptive perception (used for geometry, obstacles, and coarse terrain classification), incorporate direct physical/proprioceptive measurements, wheel slip ratio (commanded vs. actual wheel-ground speed), wheel sinkage depth (suspension deflection or visual wheel-track depth), drive-motor torque/current, and IMU-sensed body pitch/roll, so that soil cohesion, friction angle, and sinkage exponent can be inferred from the rover's actual physical interaction with the ground, not only from its visual appearance.
- Formulate a coherent risk measure: integrate Conditional Value-at-Risk (CVaR) into the MPC cost function to penalise the expected value of worst-case sinkage outcomes rather than only their mean, using the physically-grounded terrain estimate above as its input rather than a geometry-only risk map.
- Design the CVaR-SMPC controller: construct a risk-sensitive stochastic MPC that jointly optimises steering and velocity/torque commands over a finite prediction horizon, subject to hard actuator constraints (motor torque limits) and a tunable risk-bounded chance constraint on catastrophic sinkage probability.
- Benchmark in simulation: evaluate entrapment rate versus path efficiency for the proposed CVaR-SMPC against (i) a deterministic Nonlinear MPC (NMPC) baseline and (ii) a Gaussian-noise SMPC baseline, on a stochastically generated terrain map with patchy high-risk zones, and specifically test whether the added

physical sensing allows the rover to safely traverse and explore more of the visually-ambiguous terrain than a LiDAR/camera-only baseline, without increasing entrapment rate.

4. Literature Survey

The proposal draws on three intersecting bodies of work: predictive control for rover and off-road trajectory tracking, terramechanics-based slip-sinkage modelling, and CVaR-based risk-sensitive planning and control. Representative works and the corresponding gap this project targets are summarised below.

4.1 Predictive Control for Rover / Off-Road Trajectory Tracking

Zheng and Zhu [1] developed a stochastic time-varying MPC for trajectory tracking of a wheeled mobile robot, showing that explicitly modelling time-varying uncertainty in the tracking-error dynamics improves robustness over a fixed-noise-model MPC. Li, Du, and Zhang [2] surveyed path-planning approaches for autonomous vehicles specifically in off-road scenarios, highlighting that most off-road planners still rely on geometric or cost-map heuristics, i.e., LiDAR/camera-derived elevation and roughness, rather than a probabilistic, physically-grounded terrain model embedded in the controller itself. He, Sun, Yang, and Gao [3] designed an MPC-based trajectory-tracking controller for a novel wheeled-legged planetary rover, demonstrating that MPC is a practical on-board control architecture for rovers with complex mobility mechanisms, but under a deterministic terrain assumption. Taken together, this body of work establishes MPC as the standard trajectory-tracking framework for wheeled and rover platforms, yet none of these three formulations treats the terrain interaction itself as an uncertain, risk-bearing quantity inside the optimization, and none incorporates physical (proprioceptive) sensing of soil behaviour.

4.2 Terramechanics-Based Slip-Sinkage Modelling

Ishigami, Miwa, Nagatani, and Yoshida [4] derived a terramechanics-based steering model for planetary rovers on loose soil, extending Bekker-Wong theory to predict slip and lateral force during steering manoeuvres. Ding, Gao, Deng, and Tao [5] developed a wheel slip-sinkage prediction model for lunar rovers, refining the sinkage exponent to capture how sinkage grows disproportionately with slip ratio rather than remaining a static function of vertical load. Both works provide accurate, physically grounded slip-sinkage models built from exactly the kind of physical quantities (slip, torque, sinkage) this project proposes to sense onboard, but both are used for state estimation and reactive traction control rather than being embedded as the uncertain dynamics inside a predictive planner: the model informs the controller only after slip has begun, not before the rover commits to a hazardous region, and neither work couples this physical sensing with a forward-looking, risk-sensitive optimization.

4.3 CVaR and Risk-Sensitive Predictive Control

Blackmore, Ono, and Williams [6] formulated chance-constrained optimal path planning with obstacles, establishing the standard probabilistic-constraint baseline against which CVaR-based methods are typically compared; chance constraints bound only the probability of a violation, not its severity, so they cannot distinguish a barely-exceeded constraint from a catastrophic one. Chow, Tamar, Mannor, and Pavone [7] provided the foundational CVaR-based framework for risk-sensitive and robust sequential decision-making, establishing CVaR as a coherent risk measure that can be optimized via convex reformulation, the theoretical basis this project

builds its cost formulation on. Wang, Schuurmans, and Patrinos [8] proposed a sequential convexification algorithm for risk-sensitive, interaction-aware MPC with decision-dependent uncertainty, showing that a nonconvex risk-sensitive MPC problem can be solved efficiently as a sequence of convex problems, directly relevant to keeping a CVaR-SMPC solvable in real time on rover-class hardware.

Closest to this proposal, Moon, Seong, Shim, Kim, Cho, and Lee [9] proposed a sampling-based MPC with CVaR-based risk assessment for off-road environments, building a recursive CVaR risk map from elevation data and demonstrating improved success rate and smoother motion over hybrid A* and standard MPC baselines on real ground-robot trials. This is the nearest prior work to the present proposal: it confirms that CVaR-weighted risk maps meaningfully improve off-road MPC safety. However, consistent with the sensing gap identified in Section 2, its risk map is built from terrain elevation data alone (i.e., LiDAR/camera geometry), and it does not model the underlying slip-sinkage dynamics that actually cause entrapment, nor does it draw on any physical/proprioceptive measurement of the rover's own wheel-soil interaction. It uses CVaR to reshape the sampling distribution of a path-integral planner from geometric risk, rather than to constrain a terramechanics-based dynamic model informed by physical sensing.

The specific gap this project targets is therefore twofold: first, to replace or supplement LiDAR/camera-only, geometry-based risk estimation with direct physical measurement of wheel slip, sinkage, torque, and attitude, feeding a Bekker-Wong stochastic slip-sinkage model; and second, to couple that physically-grounded model directly with a CVaR-constrained SMPC optimization, so that the tail-risk of catastrophic sinkage, evaluated from actual soil-mechanical behaviour rather than surface appearance, is what shapes the rover's trajectory. This is expected to allow the controller to safely traverse and explore terrain that a geometry-only risk map would conservatively avoid, while remaining appropriately cautious in genuinely hazardous zones.

5. Proposed Approach: From LiDAR/Camera to Physically-Grounded Risk (Initial Solution)

The starting point for the controller is standard exteroceptive perception, LiDAR and camera data, used, as in prior work, for obstacle geometry and a coarse, appearance-based terrain classification (e.g., “sand,” “gravel,” “rock”). This alone, however, only estimates what the terrain looks like, not how it will mechanically behave under load. To close that gap, the following physical quantities are proposed as additional inputs, each already accessible from standard rover instrumentation or directly readable from the simulation environment:

- Wheel slip ratio, computed by comparing commanded wheel angular velocity against actual ground speed (from visual odometry or wheel encoders); large, persistent mismatch indicates loss of traction consistent with soft or loose soil.
- Wheel sinkage depth, measured via suspension deflection, or, in simulation, read directly from the physics engine's wheel-ground contact state.
- Drive-motor torque / current, elevated torque at low forward speed indicates the wheel is working against soil resistance rather than rolling freely, a classic terramechanics signature of sinkage in progress.
- IMU-sensed body pitch, roll, and vibration, detects asymmetric sinkage (e.g., one side of the rover sinking faster than the other) before it becomes visually obvious.

These four signals are precisely the quantities that the Bekker-Wong equations relate to soil cohesion, internal friction angle, and sinkage exponent: cohesion and friction angle physically cause a given slip/sinkage/torque response, so observing the response allows the soil parameters, and hence the risk of further, worse sinkage, to be inferred, rather than merely guessed from appearance.

Given the course timeline, the physical-sensing pipeline is proposed as an incremental, three-stage build, matching the project's three scheduled presentations:

- Stage 1 (for the first presentation): known/assumed soil parameters are assigned per terrain patch from published terramechanics tables, to validate the CVaR-SMPC formulation and controller in isolation from the sensing problem.
- Stage 2: coarse LiDAR/camera-based terrain classification is added, mapping each visual class to a parameter range (with uncertainty) drawn from the literature, so the controller must reason about imperfect, appearance-only knowledge, reproducing the limitation of prior work such as [9] as an explicit baseline.
- Stage 3: the physical/proprioceptive signals above are added and fused with a recursive online estimator, so that soil parameters, and the resulting CVaR-bounded risk, are updated in situ as the rover drives, rather than assumed or classified from appearance alone.

This staged approach directly supports the benchmarking objective in Section 3: at Stage 2, the controller is effectively equivalent in sensing to prior LiDAR/camera-only risk-aware planners, providing a fair baseline; at Stage 3, the added physical sensing is what should allow the rover to distinguish genuinely hazardous terrain from merely visually-ambiguous terrain, and therefore travel and explore more of the map safely rather than defaulting to conservative avoidance.

6. Expected Outcomes

- A mathematical formulation of the CVaR-SMPC controller specifically tailored to terramechanics-based slip-sinkage dynamics.
- A functional software simulation (Python/MATLAB) demonstrating rover navigation over a stochastically generated terrain map with patchy high-risk zones.
- A working online estimation pipeline that infers Bekker-Wong soil parameters from wheel slip, sinkage, torque, and IMU data, in addition to the baseline LiDAR/camera terrain classification.
- Quantitative comparison of entrapment rate and mission-completion time against deterministic NMPC and Gaussian SMPC baselines, and specifically against a LiDAR/camera-only risk-map baseline (Stage 2), to isolate the benefit of the added physical sensing.
- Visual analysis (trajectory maps / phase plots) showing that the prediction horizon actively routes the rover around zones of high tail-risk, and, where physical sensing confirms terrain is safer than it appears, through visually-ambiguous zones that a geometry-only planner would have avoided.

References

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